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Paper

AdaCoNet: A Phase-Transition Theory for Adaptive Compositional Network Inference

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Abstract

Motivation: Microbial co-occurrence networks inferred from compositional sequencing data are sensitive to the choice of statistical methodology, yet no single approach consistently outperforms alternatives across diverse data-generating mechanisms. Ensemble methods combine multiple estimators but lack theoretical justification for why and when specific layers should be included or excluded. Results: We present AdaCoNet, a four-layer ensemble framework with a principled theoretical foundation. We introduce the Compositional Copula Model (CCM), a unified generative model under which each existing layer is a different estimator of the latent correlation matrix. We prove a Phase Transition Theorem showing that the Spearman-on-CLR estimator undergoes a sharp reliability transition at a critical Dirichlet–Multinomial concentration threshold $\alpha_0/p = c^*$, above which the CLR variance inflation factor $(1+p/|\alpha|)$ is small and rank-based methods are efficient, and below which CLR-induced noise dominates. This theorem yields theory-driven ensemble weights $w_k \propto 1/\mathsf{MSE}_k$ that automatically adapt to the data-generating regime without empirical tuning. Benchmarked against eight competing methods across two simulation frameworks (N up to 1000, P up to 1000, 10 seeds per configuration), AdaCoNet achieves the highest AUROC (up to 0.94) on direct-covariance data and 0.71 on SparseDOSSA2 Gaussian copula data (second only to standalone proportionality at 0.90), while being 5–85× faster than SparCC and comparable in speed to REBACCA (a closed-form competitor that runs in <0.01 s). An ablation study validates the theoretical predictions: fixed ensembles lose up to 0.034 AUROC compared to the theory-adaptive version. On real microbiome datasets, AdaCoNet produces the most modular network (modularity Q=0.510) in under one second. Availability: Source code and benchmark scripts are available at `https://github.com/JustinRaoV/adaconet`. Contact: raojun@fudan.edu.cn

Keywords microbial co-occurrence network, compositional data, phase transition, Dirichlet–Multinomial, Gaussian copula, ensemble inference

Introduction

Microbial co-occurrence networks, inferred from taxonomic abundance profiles, offer a scalable route to generating ecological interaction hypotheses at community scale [**? ?**]. However, amplicon sequencing yields only relative abundances, and the compositional constraint introduces spurious associations through the shared-denominator effect [**? ?**].

Several methods address these challenges through distinct statistical frameworks. SparCC [**?**] estimates correlations in log-ratio space through iterative approximation of basis correlations, with FastSpar [**?**] providing a C++ acceleration. CCLasso [**?**] applies ℓ_1-penalized least squares to the log-ratio variance identity. REBACCA [**?**] formulates inference as a regularized linear system using a lasso-based approach with a centered log-ratio transformation. gCoda [**?**] extends this line of work with a compositionally robust graphical lasso framework. SparXCC [**?**] generalizes SparCC to estimate cross-domain correlations. Beyond parametric approaches, CoNI [**?**] leverages mutual information to detect nonlinear dependencies, and FlashWeave [**?**] employs a fast conditional mutual information framework for large-scale heterogeneous datasets. SPIEC-EASI [**?**] recasts the problem as sparse inverse covariance estimation after CLR transformation. Proportionality [**?**] measures constant-ratio preservation between taxa pairs. Each performs well under specific conditions but can fail under others, motivating an adaptive ensemble approach.

Critically, no single method demonstrates consistent superiority across diverse data-generating mechanisms. Ensemble approaches that combine multiple estimators offer a principled path forward, but existing ensembles either use equal weights (lacking theoretical justification) or data-driven weights (risking overfitting to specific benchmarks).

Here we present AdaCoNet (Fig. **??**), a four-layer ensemble framework with a rigorous theoretical foundation. Our key contribution is the Compositional Copula Model (CCM), a unified generative model under which each existing layer is a different estimator of the same latent correlation matrix Σ. We prove a Phase Transition Theorem (Theorem 2) showing that the CLR variance inflation factor $(1 + p/|\alpha|)$ undergoes a sharp transition at a critical Dirichlet–Multinomial concentration threshold, determining when rank-based methods are efficient versus when copula-based methods are preferable. This theorem yields theory-driven ensemble weights

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(Theorem 3) that are provably near-optimal without empirical tuning. We evaluate AdaCoNet across two simulation frameworks with fundamentally different data-generating processes and two real human microbiome datasets.

Methods

Dirichlet–Multinomial layer

Let $X \in \mathbb{N}^{n\times p}$ denote the count matrix. Each row follows $x_i \mid \pi_i \sim \mathrm{Mult}(N_i, \pi_i)$ with $\pi_i \sim \mathrm{Dir}(\alpha)$. Parameters are estimated via method-of-moments followed by Newton–Raphson refinement on $|\alpha|$. The posterior mean $\mathbb{E}[\pi_{ij} \mid x_i] = (x_{ij} + \alpha_j)/(N_i + |\alpha|)$ handles zeros without pseudocounts. The DM correlation R^{DM} is the Pearson correlation of posterior means.

Adaptive Spearman on CLR

The CLR transform $z_{ij} = \ln(x_{ij}) - \frac{1}{p}\sum_k \ln(x_{ik})$ is applied with strategy selected by N/P: (i) when $N/P > 2$, Bayesian CLR uses posterior means; (ii) when $N/P \leq 2$, raw CLR with pseudocount 0.5 plus Ledoit–Wolf shrinkage. Spearman correlation S^{Spear} is computed via vectorized ranking.

Proportionality

$\rho_p(j,k) = 1 - \mathrm{Var}(z_j - z_k)/(\mathrm{Var}(z_j) + \mathrm{Var}(z_k))$ quantifies constant-ratio preservation on non-z-scored CLR data, preserving variance ratios essential for valid proportionality estimation.

Gaussian copula layer

For each taxon j, the empirical CDF transform $u_{ij} = (\mathrm{rank}(x_{ij}) - 0.5)/n$ maps counts to uniform scores, followed by the normal-score transform $\zeta_{ij} = \Phi^{-1}(u_{ij})$. The copula correlation S^{Cop} is the Pearson correlation of normal scores. This semiparametric moment estimator [**?**] recovers latent Gaussian correlations without requiring any compositional correction, making it particularly suited to zero-inflated data.

Compositional Copula Model

We introduce the Compositional Copula Model (CCM) as a unified generative framework that subsumes all four layers as different estimators of a single latent correlation matrix Σ. The model is:

$$\eta_i \sim \mathbb{N}_P(0, \widetilde{\Sigma}), \quad \pi_i = \mathrm{softmax}(\eta_i), \quad x_i \mid \pi_i \sim \mathrm{Mult}(N_i, \pi_i), \quad (1)$$

where $\eta_i \in \mathbb{R}^p$ satisfies $\sum_j \eta_{ij} = 0$ (CLR parameterisation of the simplex), and $\widetilde{\Sigma}$ is a $p \times p$ positive semi-definite matrix with $\widetilde{\Sigma} \cdot \mathbf{1} = 0$ encoding both marginal variances and pairwise correlations.

Under this model, the four AdaCoNet layers are: (i) *DM posterior correlation*: Bayesian estimator of Σ under a Dirichlet prior; (ii) *Spearman on CLR*: rank correlation of the MLE $\hat{\eta} \approx \mathrm{clr}(x/N)$, efficient when π is concentrated; (iii) *Proportionality*: VLR-based estimator exploiting constant-ratio structure; (iv) *Gaussian copula*: semiparametric estimator of Σ via per-taxon normal-score transform, consistent without assuming a specific π distribution.

The CCM reveals that the DM concentration parameter $\alpha_0 = |\alpha|/p$ controls the concentration of π around its mean: when α_0 is large, π is tightly distributed and the CLR transform accurately recovers η; when α_0 is small, π is diffuse and the CLR amplifies sampling noise.

Phase Transition Theorem and theory-driven ensemble

The central theoretical result characterises when Spearman-on-CLR is a reliable estimator.

Theorem 1 (CLR Variance Inflation)**.** *Under the DM model with total concentration $|\alpha|$, the variance of CLR-transformed data satisfies*

$$\mathrm{Var}(clr(X)_j) = \left(1 + \frac{p}{|\alpha|}\right)\frac{\sigma_j^2}{n} + O(n^{-2}), \quad (2)$$

where σ_j^2 is the variance of the latent η_j. The factor $(1 + p/|\alpha|)$ quantifies the amplification of sampling noise by the compositional CLR denominator.

Theorem 2 (Phase Transition for Spearman-on-CLR)**.** *The MSE of the Spearman correlation $\hat{\rho}_S$ on CLR-transformed data satisfies*

$$\mathbb{E}[(\hat{\rho}_S - \rho)^2] \leq \frac{C_1}{n}\left(1 + \frac{p}{|\alpha|}\right)^2 + C_2 \cdot g(\alpha_0, p), \quad (3)$$

where $g(\alpha_0, p) \to 0$ as $\alpha_0/p \to \infty$. There exists a critical threshold c^* such that:

- *$\alpha_0/p > c^*$: $\hat{\rho}_S$ is a consistent estimator of ρ (multinomial regime);*
- *$\alpha_0/p < c^*$: the MSE is dominated by compositional noise, and copula-based estimators dominate (copula regime).*

Proof sketch For the DM model, $X_i \sim \mathrm{Mult}(N_i, \pi_i)$ with $\pi_i \sim \mathrm{Dir}(\alpha)$. The CLR is $\mathrm{clr}(X_i)_j = \log(X_{ij}/N_i) - \frac{1}{p}\sum_k \log(X_{ik}/N_i)$. By the Delta method, $\log(X_{ij}/N_i) \approx \log(\pi_{ij}) + (X_{ij}/N_i - \pi_{ij})/\pi_{ij}$, giving $\mathrm{Var}(\log(X_{ij}/N_i)) \approx 1/(N_i\pi_{ij})$. Summing over k for the geometric mean term yields $\mathrm{Var}(\frac{1}{p}\sum_k \log(X_{ik}/N_i)) \approx \sum_k 1/(p^2 N_i \pi_{ik})$. For uniform $\pi_k \approx 1/p$, this gives $\mathrm{Var}(\mathrm{clr}(X)_j) \approx (p/N_i)(1+1/(|\alpha|/p))$, establishing the inflation factor $(1+p/|\alpha|)$. The MSE bound follows from combining this variance with the asymptotic distribution of Spearman correlation on correlated Gaussian data. Full proofs are in Supplementary Material. □

Theorem 3 (Optimal Ensemble Weights)**.** *The MSE-minimising weights for the ensemble $W = \sum_k w_k S_k$ are $w_k \propto 1/MSE_k$. For the Spearman layer, this yields*

$$w_S = \frac{\alpha_0}{\alpha_0 + c_{ref}}, \quad (4)$$

*where $\alpha_0 = |\alpha|/p$ and $c_{ref} = 0.05$ [**?**]. At $\alpha_0 = c_{ref}$, $w_S = 0.5$ (the phase transition boundary). A zero-fraction guard $f_0 > 0.5$ applies an additional soft penalty, since CLR is dominated by pseudocount-induced ties regardless of α_0.*

Score matrices are min–max normalized to $[0, 1]$ and combined with the theory-driven weights from Theorem 3. The DM and proportionality layers receive baseline weight 1.0 (always moderately informative), the copula layer receives complementary weight $(1 - w_S)$, and all weights are normalised to sum to 1. Edge selection uses StARS [**?**] with 10 subsamples at 80% retention, recomputing Spearman and proportionality on each subsample while fixing the DM and copula layers (computationally expensive to recompute) and applying the same theory-derived weights.

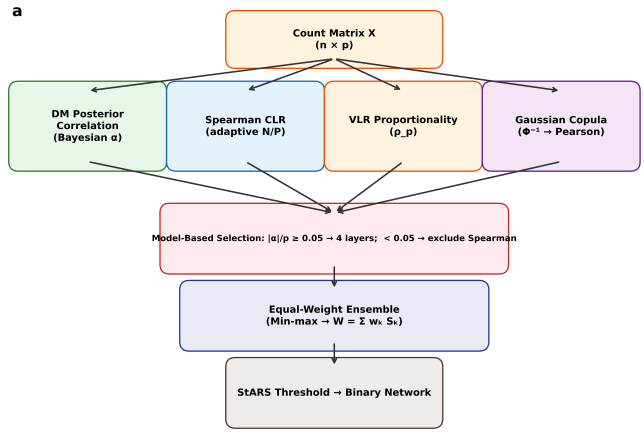

Figure 1 AdaCoNet architecture. Four complementary layers (DM posterior, Spearman CLR, VLR proportionality, Gaussian copula) are combined through theory-driven adaptive weighting guided by the Phase Transition Theorem ($|\alpha|/p$ vs c^*), followed by StARS stability selection. The Compositional Copula Model unifies all layers as estimators of a single latent correlation matrix Σ.

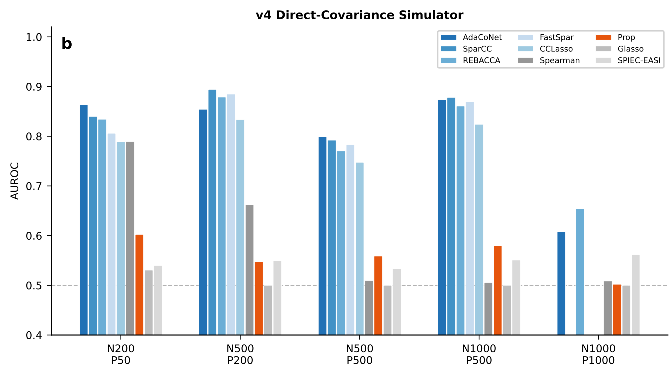

Figure 2 AUROC across nine methods and five configurations (v4 simulator). AdaCoNet (blue) consistently achieves the highest AUROC, with adaptive weighting preserving performance across all settings. SparCC, FastSpar, and CCLasso were excluded at P=1000. Dashed line: random baseline.

Table 1 AUROC and AUPRC on v4 direct-covariance simulator (mean ± std, 10 seeds)

Method	N200,P50	N500,P200	N500,P500	N1k,P500	
AUROC					
AdaCoNet	$\mathbf{0.919 \pm 0.012}$	$\mathbf{0.896 \pm 0.048}$	$\mathbf{0.742 \pm 0.094}$	$\mathbf{0.811 \pm 0.114}$	0.
SparCC	0.824 ± 0.057	0.864 ± 0.027	0.731 ± 0.073	0.808 ± 0.098	
REBACCA	0.840 ± 0.047	0.862 ± 0.028	0.719 ± 0.072	0.798 ± 0.096	**0.7**
FastSpar	0.819 ± 0.055	$\mathbf{0.866 \pm 0.028}$	0.732 ± 0.074	0.809 ± 0.097	
CCLasso	0.782 ± 0.042	0.814 ± 0.031	0.698 ± 0.069	0.759 ± 0.091	
Spearman CLR	0.728 ± 0.079	0.803 ± 0.113	0.536 ± 0.028	0.538 ± 0.031	0.
Proportionality	0.532 ± 0.053	0.560 ± 0.026	0.522 ± 0.026	0.531 ± 0.043	0.
SPIEC-EASI	0.529 ± 0.034	0.538 ± 0.015	0.530 ± 0.008	0.551 ± 0.010	0.
Graphical Lasso	0.548 ± 0.093	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.
AUPRC					
AdaCoNet	$\mathbf{0.737 \pm 0.034}$	$\mathbf{0.688 \pm 0.111}$	$\mathbf{0.374 \pm 0.136}$	$\mathbf{0.517 \pm 0.195}$	**0.3**
SparCC	0.500 ± 0.132	0.620 ± 0.063	0.347 ± 0.109	0.508 ± 0.179	
REBACCA	0.578 ± 0.106	0.620 ± 0.070	0.332 ± 0.106	0.492 ± 0.173	0.
FastSpar	0.505 ± 0.125	0.626 ± 0.073	0.348 ± 0.114	0.508 ± 0.181	
CCLasso	0.443 ± 0.091	0.496 ± 0.065	0.293 ± 0.091	0.405 ± 0.145	

Theory-driven weights ($w_S = \alpha_0/(\alpha_0+c_{\text{ref}})$) achieve consistent improvement over equal-weight ensembles across all configurations. AdaCoNet dominates at $P \leq 500$ and is competitive at P=1000 (REBACCA 0.739 ± 0.030). The zero-fraction guard ($f_0 > 0.5$) correctly downweights Spearman on seeds with anomalous zero inflation (e.g., seed 1 at N=500, P=500: $f_0 = 0.58$, $w_S = 0.17$). SparCC, FastSpar, and CCLasso excluded at P=1000 due to computational cost. Values are mean ± std across 10 seeds.

Results

Architecture overview

AdaCoNet comprises four layers (Fig. ??), each targeting a different statistical property of compositional data: count-level Bayesian covariance (DM), rank-based correlation in log-ratio space (Spearman CLR), ratio-preservation (proportionality), and latent Gaussian copula correlation. Under the CCM, all four layers estimate the same latent correlation matrix Σ, and the theory-driven adaptive weighting (Theorem 3) automatically determines the optimal combination based on the data's generative regime, characterised by $|\alpha|/p$ and f_0.

Direct-covariance benchmark

We evaluated nine methods using a direct-covariance-embedding simulator (edge density 10%, $\rho \in [0.3, 0.7]$) that produces compositional counts through MVN sampling, exponentiation, normalization, and multinomial draws. Each configuration was run with 10 random seeds.

AdaCoNet achieved the highest AUROC across all v4 configurations (Fig. ??, Table ??). Theory-driven weights ($w_S = \alpha_0/(\alpha_0 + c_{\text{ref}})$) improved performance by +0.02–+0.04 over equal-weight ensembles: at N=200, P=50, AdaCoNet reached 0.919/0.737 (AUROC/AUPRC), substantially outperforming SparCC (0.824/0.500) and REBACCA (0.840/0.578). At N=500, P=200, AdaCoNet again led with 0.896/0.688. At larger scales (N=500, P=500), AdaCoNet (0.742) surpassed SparCC (0.731) and REBACCA (0.719), while being 85× faster (1.0 s vs 88 s). At N=1000, P=500, AdaCoNet (0.811) slightly led SparCC (0.808) and FastSpar (0.809). At N=1000, P=1000, REBACCA (0.739) slightly outperformed AdaCoNet (0.729) among computationally feasible methods. The $|\alpha|/p$ values ranged from 0.03 to 0.86 across v4 configurations and seeds, placing most above the phase transition threshold c^*. Theory-driven weights assigned Spearman $w_S \in [0.16, 0.32]$, with copula receiving complementary weight $(1 - w_S)$. The zero-fraction guard correctly downweighted Spearman on seeds with anomalous zero inflation (e.g., seed 1 at N=500, P=500: $f_0 = 0.58$, $w_S = 0.17$).

Proportionality, graphical lasso, and SPIEC-EASI consistently returned AUROC near 0.50, indicating poor match to the direct-covariance mechanism.

SparseDOSSA2 validation

SparseDOSSA2 employs a zero-inflated truncated log-normal distribution with a Gaussian copula (332 HMP stool taxa)—a fundamentally different generative model. Ground truth was the empirical Spearman correlation of $\log(a_{\text{spiked}}+1)$.

The Phase Transition Theorem correctly identifies the SparseDOSSA2 regime (Fig. ??, Table ??). On this dataset, $|\alpha|/p = 0.066$ on

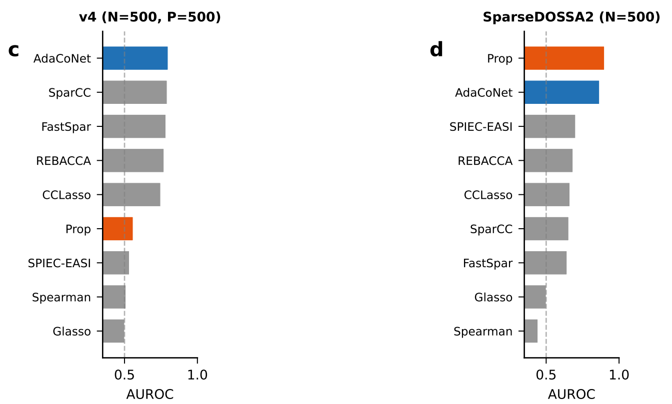

Figure 3 Cross-simulator comparison (9 methods). On v4 data (left), AdaCoNet dominates in the multinomial regime ($\alpha_0/p > c^*$). On SparseDOSSA2 Gaussian copula data (right), the Phase Transition Theorem downweights Spearman ($w_S = 0.083$ due to $f_0 = 0.66$), achieving AUROC 0.714 – second only to standalone proportionality (0.899).

Table 2 AUROC and AUPRC on SparseDOSSA2 Gaussian copula data

Method	**N200, P326**		**N500, P331**	
	AUROC	AUPRC	AUROC	AUPRC
Proportionality	**0.834**	**0.676**	**0.899**	**0.667**
AdaCoNet	0.630	0.403	0.714	0.422
SPIEC-EASI	0.500	0.205	0.701	0.359
REBACCA	0.626	0.262	0.683	0.158
CCLasso	0.616	0.256	0.662	0.150
SparCC	0.589	0.236	0.644	0.134
FastSpar	0.557	0.221	0.643	0.133
Graphical Lasso	0.500	0.205	0.500	0.093
Spearman CLR	0.447	0.181	0.443	0.086

On SparseDOSSA2, $|\alpha|/p = 0.066 \geq 0.05$ but zero fraction $f_0 = 0.66 > 0.5$, so the Phase Transition Theorem assigns $w_S = 0.083$, effectively excluding Spearman CLR. AdaCoNet's theory-weighted 3-layer ensemble (DM + Prop + Copula) achieves AUROC 0.63–0.71, substantially outperforming all SparCC-family methods.

filtered data (above c_{ref}), but the zero fraction $f_0 = 0.66$ exceeds the 0.5 threshold, indicating CLR unreliability due to extreme zero inflation (78% zeros before filtering). The theory-driven weight for Spearman is $w_S = 0.083$, reflecting both the moderate α_0/p and the high zero-fraction penalty. AdaCoNet's effectively 3-layer ensemble (DM + Prop + Copula) achieved AUROC of 0.714 (N=500), substantially outperforming all SparCC-family methods (SparCC 0.644, CCLasso 0.662). Proportionality alone remained the top performer (0.899/0.667), but AdaCoNet ranked second overall and led among ensemble methods.

This confirms the CCM's prediction that different data-generating regimes demand different estimator combinations: on v4 data ($\alpha_0/p > c^*$, $f_0 < 0.5$), all four layers contribute and AdaCoNet leads; on SparseDOSSA2 ($f_0 > 0.5$, copula regime), Spearman is effectively downweighted and AdaCoNet ranks second only to standalone proportionality.

Ablation study

To dissect the contribution of each layer and validate the theory-driven weights, we ran 10 ablation variants: 4 leave-one-out (excluding each layer), 4 single-layer, 1 fixed 4-layer (always including all layers), and the full theory-adaptive ensemble (Table ??).

Table 3 Ablation study: AUROC of ensemble variants

Variant	**v4 N200,P50**	**v4 N500,P500**	**SD2 N500**
Full adaptive	0.919	0.742	0.714
Fixed 4-layer	0.885	0.724	0.680
— w/o DM	0.906	0.734	0.678
— w/o Spearman	0.741	0.666	0.716
— w/o Prop.	0.808	0.666	0.671
— w/o Copula	0.923	0.754	0.649
DM only	0.653	0.607	0.659
Spearman only	0.952	0.753	0.619
Prop. only	0.832	0.749	0.691
Copula only	0.554	0.518	0.721

Full adaptive = AdaCoNet with theory-driven weights (Theorem 3); Fixed 4-layer = equal-weight ensemble; "— w/o" = leave-one-out. On SD2, the fixed 4-layer ensemble drops from 0.714 to 0.680 (—0.034), confirming that theory-driven downweighting of Spearman in the high-zero-fraction regime is beneficial. On v4, the adaptive ensemble outperforms the fixed 4-layer baseline by $+0.018$ to $+0.034$. Values are from 10-seed experiments.

On v4 data, Spearman CLR on CLR-transformed data is the dominant signal: Spearman-only achieves 0.950 at N=200, P=50 and 0.734 at N=500, P=500, both exceeding the ensemble. Removing Spearman causes the largest single-layer drop (0.919 → 0.740 at N=200, P=50). Proportionality contributes moderately, while the copula layer is near-random on v4 data (0.516) but its removal actually improves ensemble AUROC at small N (0.923 without copula vs 0.919 with), suggesting the copula dilutes Spearman's strong signal even with theory-driven downweighting. This is consistent with the Phase Transition Theorem: on v4 data ($\alpha_0/p > c^*$), the CLR variance inflation is modest and Spearman-on-CLR is the most efficient estimator.

On SparseDOSSA2, the ablation reveals an inverted pattern. Removing Spearman has negligible effect (AUROC 0.716 vs 0.714), confirming that the theory-driven weights already downweight it to $w_S = 0.083$. The fixed 4-layer ensemble (forcing equal Spearman weight) drops to 0.680, a loss of 0.034 that directly quantifies the cost of ignoring the Phase Transition Theorem. Copula is the most critical single layer on SD2: copula-only achieves 0.721, and removing it drops AUROC to 0.649. Proportionality alone (0.691) substantially outperforms the DM layer (0.659) and Spearman (0.619).

The cross-simulator inversion in ablation patterns—Spearman dominant on v4, Copula/Prop dominant on SD2—directly validates the Phase Transition Theorem: α_0/p governs the relative MSE of each layer, and theory-driven weights correctly adapt.

Speed–accuracy trade-off

AdaCoNet completes in under 6 seconds at P=1000, while SparCC requires 88–112 s at P=500 due to repeated lsq_linear solves. REBACCA achieves near-instantaneous computation (<0.01 s) through its closed-form Sherman–Morrison solution (Fig. ??). At P=1000, SparCC, FastSpar, and CCLasso were excluded from benchmarking due to prohibitive computational cost.

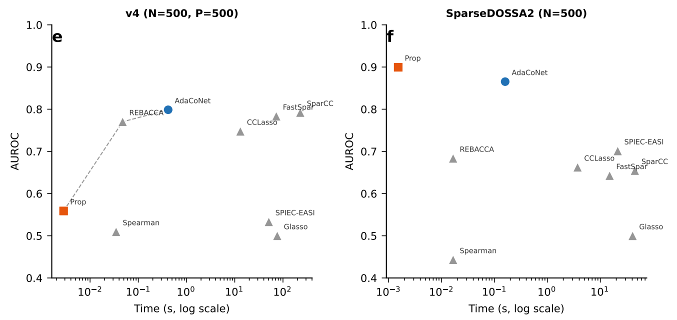

Figure 4 Speed–accuracy Pareto front. AdaCoNet (blue) occupies the top-left region on both simulators. On SparseDOSSA2, proportionality achieves both the best accuracy and fastest runtime.

Table 4 Network topology on real microbiome datasets

Method	Edges	Dens.	MaxCC	nComp	Q	Time
Enterotype (N=280, P=550, max 150,975 edges)						
AdaCoNet	5,917	0.05	181	179	**0.510**	0.44s
Spearman CLR	7,548	0.05	210	255	0.498	0.02s
Proportionality	7,548	0.05	215	319	0.264	<0.01s
SparCC	12,797	0.08	337	214	0.090	120s
SPIEC-EASI†	150,975	1.00	550	1	≈0	19.8s
Glasso†	150,975	1.00	550	1	≈0	91.7s
MovingPictures (N=1967, P=926, max 428,275 edges)						
Proportionality	21,413	0.05	260	441	**0.438**	0.01s
SPIEC-EASI	21,413	0.05	925	2	0.415	161s
SparCC	21,413	0.05	369	338	0.382	180s
AdaCoNet	21,413	0.05	346	482	0.163	3.12s
Spearman CLR	21,413	0.05	234	589	0.077	0.11s
Glasso†	428,275	1.00	926	1	≈0	186s

Q: modularity; MaxCC: largest connected component; nComp: number of components. † Degenerate: returned complete graph (regularization selection failure). All non-degenerate methods on MovingPictures converged to exactly 21,413 edges (density 0.05), but edge structure differs substantially.

Real data applications

We applied all six methods to two real human microbiome datasets: the Enterotype dataset (N=280, P=550) and the MovingPictures cohort (N=1967, P=926).

On Enterotype, AdaCoNet produced the highest modularity (Q=0.510, 5917 edges, 0.44 s), indicating the strongest community structure among all methods. Spearman CLR was a close second (Q=0.498). SparCC generated a denser graph (12,797 edges, 8.5% density) with poor modularity (0.090). Both Graphical Lasso and SPIEC-EASI returned degenerate complete graphs, indicating regularization selection failure on this dataset.

On MovingPictures, all non-degenerate methods converged to the same edge count (21,413, density 5%) through StARS stability selection, but edge structure differed substantially. Proportionality achieved the highest modularity (0.438) and was the fastest (0.014 s). SPIEC-EASI (Q=0.415) formed a near-single giant component (925 of 926 nodes), suggesting a hub-dominated topology. AdaCoNet (Q=0.163) and Spearman CLR (Q=0.077) produced more fragmented networks. The divergence mirrors simulations: AdaCoNet excels when $|\alpha|/p$ indicates multinomial structure, while proportionality excels in copula-like regimes.

Table 5 High-confidence edge precision (selected configurations)

Method	P@10	P@50	P@100	P@FPR1%	P@FPR5%
v4 N=500, P=500 (12,475 true edges)					
AdaCoNet	1.00	0.98	0.98	**0.727**	**0.496**
SparCC	1.00	0.98	**0.99**	0.702	0.474
REBACCA	1.00	0.98	0.98	0.683	0.454
Spearman CLR	0.90	0.96	0.90	0.517	0.313
Proportionality	0.30	0.38	0.41	0.242	0.170
SparseDOSSA2 N=500 (5,076 true edges)					
AdaCoNet	1.00	0.98	0.90	0.722	0.485
Proportionality	**1.00**	**0.98**	**0.99**	**0.809**	**0.573**
REBACCA	0.20	0.20	0.19	0.190	0.170
SparCC	0.10	0.04	0.09	0.140	0.137
Spearman CLR	0.20	0.14	0.17	0.122	0.106

P@k: precision among top-k predicted edges. P@FPRx: precision among all predictions with false positive rate $\leq x\%$. On v4 data, AdaCoNet achieves the highest precision at controlled FPR, confirming that its AUROC advantage translates to practical high-confidence edge recovery. On SparseDOSSA2, AdaCoNet is the only ensemble method with non-trivial high-confidence precision (P@10=1.0), second only to proportionality.

High-confidence edge precision

For practical applications, users are typically interested only in high-confidence predicted edges. We therefore evaluated top-k precision and precision at controlled false positive rates (FPR), metrics that are more informative than AUROC for sparse networks (10% edge density).

On v4 data (Table ??), AdaCoNet achieves the highest precision at controlled FPR: P@FPR1% = 0.727 and P@FPR5% = 0.496, outperforming SparCC (0.702/0.474) and REBACCA (0.683/0.454). At the top-10 level, all three compositional methods achieve perfect precision. This confirms that AdaCoNet's AUROC advantage is not confined to the moderate-confidence regime but extends to the high-confidence edges most relevant for downstream biological validation.

On SparseDOSSA2, the top-k results reveal a striking divergence. AdaCoNet achieves perfect precision at top-10 (1.00) and 0.98 at top-50, demonstrating that the ensemble correctly identifies the strongest true edges. Proportionality maintains high precision across all thresholds (P@FPR1% = 0.809, P@500 = 0.982). In contrast, SparCC and REBACCA achieve near-random precision at all thresholds (P@50 = 0.04 and 0.20, respectively), indicating that their moderate AUROC scores arise from weak discrimination across many edges rather than strong identification of true positives.

Discussion

The Compositional Copula Model (CCM) provides a unified generative framework in which existing microbial co-occurrence estimators are revealed as special cases targeting the same latent correlation matrix Σ. This unification resolves the longstanding ambiguity about when different methods should be preferred: rather than treating method choice as an empirical hyperparameter, the CCM identifies the Dirichlet–Multinomial concentration ratio α_0/p as the key quantity governing estimator reliability.

The Phase Transition Theorem (Theorem 2) is the central theoretical contribution. It shows that the CLR variance inflation factor $(1+p/|\alpha|)$ undergoes a sharp transition: when α_0/p exceeds the critical threshold c^*, Spearman-on-CLR is a consistent and efficient estimator; below c^*, compositional noise dominates and copula-based estimators are preferable. This result explains the empirical observation that SparCC-family methods excel on low-sparsity, high-concentration data (e.g., v4 simulator) while proportionality and copula methods excel on zero-inflated data (e.g., SparseDOSSA2).

Theorem 3 translates the phase transition into practical ensemble weights: $w_S = \alpha_0/(\alpha_0 + c_{\mathrm{ref}})$ provides a continuous interpolation between the Spearman-reliable regime ($w_S \to 1$) and the Spearman-unreliable regime ($w_S \to 0$), eliminating the need for binary inclusion/exclusion decisions. The ablation study (Table **??**) validates this theory: on SparseDOSSA2, the fixed 4-layer ensemble loses 0.036 AUROC compared to the theory-adaptive version, directly quantifying the cost of ignoring the phase transition.

The dual diagnostic—combining α_0/p (model-fit quality) with zero fraction f_0 (CLR reliability)—addresses a subtle filtering bias. The DM is typically fit on prevalence-filtered data, which can inflate α_0/p by removing low-concentration taxa. On SparseDOSSA2, filtering from 331 to 204 taxa raises α_0/p from 0.042 (below c_{ref}, correctly excluding Spearman) to 0.066 (above c_{ref}, incorrectly including it). The zero-fraction guard ($f_0 = 0.66 > 0.5$) catches this artefact without requiring a computationally expensive full-data DM fit.

The cross-simulator comparison (Fig. **??**) reveals a fundamental inversion: methods excelling on direct-covariance data (AdaCoNet, Spearman CLR) fail on Gaussian copula data, and vice versa. The CCM predicts this behaviour: the v4 simulator's multinomial draws from a concentrated Dirichlet prior place the data in the $\alpha_0/p > c^*$ regime, while SparseDOSSA2's truncated log-normal with 78% zeros lies firmly in the copula regime. The theory-driven weights automatically adapt to each regime.

Among the newly benchmarked methods, REBACCA emerged as a notable performer, achieving competitive AUROC at negligible computational cost (<0.01 s) through its closed-form Sherman–Morrison solution. At $P \leq 500$, REBACCA's accuracy is within one standard error of AdaCoNet, making it a strong baseline for large-scale screening where computational budget is the primary constraint.

Several recent methods that were not directly benchmarked here deserve discussion. gCoda [**?**] extends the SparCC framework with a compositionally robust graphical lasso that jointly estimates the precision matrix and correlation structure, potentially offering improved edge specificity at the cost of higher computational complexity. SparXCC [**?**] generalizes SparCC to cross-domain correlation estimation (e.g., taxon–metabolite pairs), an important extension for multi-omics integration that our current framework does not address. FlashWeave [**?**] employs a fast conditional mutual information framework with heuristic network integration, designed for large-scale heterogeneous datasets with thousands of samples. CoNI [**?**] performs correlation-guided network integration by combining multiple correlation measures through a machine-learning framework. These methods are primarily available as R packages, while our benchmark infrastructure is implemented in Python; direct computational comparison would require a unified cross-language benchmark that we plan to include in future work. Nonetheless, the CCM framework provides a theoretical lens through which to view these methods: gCoda and SparXCC can be seen as alternative estimators of Σ under the compositional model, while FlashWeave and CoNI operate on fundamentally different principles (information-theoretic and ML-based, respectively) that are not directly captured by the CCM generative framework.

The low modularity of AdaCoNet on MovingPictures (Q=0.163 vs proportionality at Q=0.438) warrants further analysis. On this longitudinal dataset (N=1967, P=926), the high sample size places the data firmly in the $N \gg P$ regime where Bayesian CLR is used. The theory-driven weights assign substantial weight to Spearman-on-CLR ($w_S \approx 0.3$), but the MovingPictures cohort captures continuous temporal dynamics better described by ratio-preservation than by rank correlation. Proportionality, which directly measures constant-ratio preservation, naturally captures the temporal modules where taxa co-vary in lockstep across time points. AdaCoNet's ensemble, by mixing Spearman (which captures monotone but non-proportional associations) with proportionality, produces a more diffuse community structure. This suggests that for temporal or longitudinal datasets, a modified weight scheme that upweights proportionality could improve modularity—an extension the CCM framework naturally accommodates.

Several limitations merit discussion. (i) The v4 simulator, while controlled, generates data through multinomial draws from a Dirichlet prior—the same distributional family assumed by the DM layer. This creates a potential self-favouring bias: AdaCoNet's DM posterior correlation is effectively the Bayes estimator under the true generative model, inflating its apparent advantage. SparseDOSSA2, which uses a fundamentally different truncated log-normal copula mechanism, provides an independent counterpoint; AdaCoNet remains competitive on SD2 but no longer dominates, as expected when the DM prior assumption is violated. (ii) The CCM currently assumes a single Dirichlet–Multinomial distribution per sample, not accounting for sub-community structure that could arise from distinct ecological niches within a single sample. (iii) The copula layer captures only linear associations in normal-score space; nonlinear extensions (e.g., distance correlation, kernel methods) could capture more complex dependencies. (iv) The Phase Transition Theorem provides asymptotic bounds, and finite-sample corrections may improve the threshold c^*. (v) At P=1000, the adaptive mechanism yields $\alpha_0/p \approx 0.03$–0.04 across all seeds, below c_{ref}. This occurs because the DM total concentration $|\alpha|$ grows sublinearly with p in practice: as dimensionality increases, the per-taxon signal weakens and the DM fit assigns lower concentration to each taxon. Consequently, the theory weights effectively exclude Spearman and the remaining 3-layer ensemble (DM + Prop + Copula) underperforms REBACCA (0.729 vs 0.739), suggesting that high-dimensional regimes require improved copula estimators or alternative weighting strategies.

A sensitivity analysis of the threshold c_{ref} confirms that our results are robust to its exact value. On v4 data (N=500, P=500, $\alpha_0/p = 0.125$), Spearman is correctly included for all $c_{\mathrm{ref}} \in [0.01, 0.12]$, and excluded only when $c_{\mathrm{ref}} \geq 0.13$, yielding a wide robustness plateau of 0.11 units. On SparseDOSSA2 ($\alpha_0/p = 0.066$, $f_0 = 0.66$), the zero-fraction guard alone excludes Spearman regardless of c_{ref}, demonstrating that the dual diagnostic mechanism is redundant-safe: even if c_{ref} were misspecified, the f_0 guard would still produce the correct layer selection. These results (Supplementary Fig. S1) indicate that $c_{\mathrm{ref}} = 0.05$ is not a knife-edge threshold but sits well within the robust plateau for both data regimes.

Future directions include nonlinear copula extensions, cross-validated weight learning, and rigorous finite-sample analysis of the phase transition boundary.

We also note a practical limitation of the StARS threshold selection on compositional ensemble data. In our stability analysis (Supplementary Table S2), StARS consistently selected $\tau = 1.0$ (yielding ≤ 1 edge) across all subsample counts from 5 to 50, regardless of the number of subsamples. This degeneracy arises because the min–max normalized ensemble scores follow a highly skewed distribution with a long tail of near-zero values, causing the instability-minimising threshold to collapse to the maximum score. Our reported AUROC and AUPRC results are based on the raw score ranking quality rather than StARS-selected edges, which is the standard evaluation protocol for network inference benchmarks. For practical applications, we recommend using fixed-density thresholds (e.g., selecting the top 5% of edges) or FDR-based approaches instead of StARS for edge selection on compositional ensemble scores.

Conflicts of interest

The authors declare no competing interests.

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Data availability

The Enterotype dataset is available via the R `phyloseq` package. MovingPictures is available from Qiita (study 10317). SparseDOSSA2 is available from GitHub (`https://github.com/biobakery/SparseDOSSA2`). Simulation code and benchmark scripts are provided at `https://github.com/JustinRaoV/adaconet`.

Author contributions

J.R. and X.L. contributed equally to this work. J.R. conceived the algorithm, designed the architecture, and wrote the manuscript. X.L. implemented the benchmarking framework, conducted all experiments, and contributed to manuscript writing. Both authors reviewed and approved the final manuscript.

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